

**College Tuition Mini-Project:**  
**How much does it cost to go to college?**

Today we will be using our understanding of recursive sequences to investigate something that will be near to our hearts very soon: student loans! In this project, you will select an in-state “safety school” and a “dream school” you would like to attend (this will likely be out of state, but both can be in-state schools so long as one is more expensive than the other). We will explore how these differences in tuition will affect your TIP (total interest paid) over the life of your student loans.

**RUBRIC: 30 TOTAL POINTS [Due: May 4<sup>th</sup>]**

On the next page, you will find an example project with the data already filled in. You will create your own tables using the tuition information you researched about your chosen universities.

1. [3 pts] Cost per Year: Research your chosen universities cost of attendance page to determine the annual cost of attendance. Be sure to include tuition, room and board, books, and other academic fees that you would be financing through your student loans.
2. [3 pts] Total cost of Attendance: Take the above number and multiply by four years. You might consider removing room and board after your freshman year if you intend to live off campus, but you should consider how you will pay for rent and other living expenses if you do not finance these costs through your student loans.
3. [3 pts] Amount financed: For this project, this will be the same as your cost of attendance. In real life, this will be your cost of attendance minus grants and scholarships.
4. [3 pts] Annual Percentage Rate: **In this project, we will assume you are receiving the average direct subsidized & unsubsidized student loan rate of 6.39%.** The interest rate for private student loans is typically higher. Remember, this is NOT the same as your annual percentage rate! You must take compounding into consideration.
5. [3 pts] Monthly Payment: **This is the dollar amount the loan will cost you. In this project we will assume your loan term is the Federal Standard Plan term of 10 years.** In real life, there are more flexible repayment options, which include extended payments terms and refinancing, but we will keep it simple for now. Use the monthly payment formula to calculate your payment:

$$M = P \frac{r(1 + r)^n}{(1 + r)^n - 1}$$

P = principal, r = rate, n = number of months (120)

6. [3 pts] Total of Payments: Take your monthly payment and multiply by 120 months. This is the total amount you will pay over the life of the loan.

7. [3 pts] Finance Charge: Take the sum of your 120 monthly payments and subtract the total amount borrowed. This is the total interest paid and can be thought of as the cost of borrowing the money.
8. [3 pts] Total Interest Percentage: Take the total interest paid as a percentage of the original borrowed amount. (Scary!)
9. [3 pts] Data Tables: In excel or google sheets, make two data tables which show your balance on months 0 to 120 calculated recursively using the techniques discussed in class.
10. [2 pt] Presentation: The report is printed out, formatted correctly, includes your name, and all boxes are in a table on a single piece of paper.

| <b>Loan Calculations</b>                                                                                                                         | <b>In State</b><br>(William & Mary) | <b>Out of State</b><br>(Sewanee) | <b>Difference</b> |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------|-------------------|
| <b>Cost per year</b><br>Estimated amount you will pay per year for tuition, books, academics fees. Exclude amounts that will not be financed.    | \$43,057                            | \$73,354                         | \$30,297          |
| <b>Total cost of Attendance</b><br>The amount above times 4 (assume you will attend college for 4 years)                                         | \$172,228                           | \$301,416                        | \$129,188         |
| <b>Amount Financed</b><br>The loan amount available after paying your upfront finance charge.                                                    | \$172,228                           | \$301,416                        | \$129,188         |
| <b>Annual Percentage Rate (APR)</b><br>Your costs over the loan term expressed as a rate. This is not your interest rate.                        | 6.58%                               | 6.58%                            | 0%                |
| <b>Monthly Payment</b><br>How much you will need to pay per month to pay off the loan within the term.                                           | \$1,946.00                          | \$3,405.67                       | \$1459.67         |
| <b>Total of Payments</b><br>Total you will have paid after you make all payments of principal, interest, and loan costs.                         | \$233,520.00                        | \$408,680.40                     | \$175,160.40      |
| <b>Finance Charge</b><br>The dollar amount the loan will cost you.                                                                               | \$61,292.00                         | \$107,264.40                     | \$45,972.40       |
| <b>Total Interest Percentage (TIP)</b><br>The total amount of interest that you will pay over the loan term as a percentage of your loan amount. | 35.6%                               | 35.6%                            | 0%                |





| Month | Sewanee<br>Balance | W&M<br>Balance | Month | Sewanee<br>Balance | W&M<br>Balance |
|-------|--------------------|----------------|-------|--------------------|----------------|
| 0     | \$ 301,416.00      | \$ 172,228.00  | 60    | \$ 174,518.73      | \$ 99,718.48   |
| 1     | \$ 299,615.37      | \$ 171,199.11  | 61    | \$ 172,042.37      | \$ 98,303.48   |
| 2     | \$ 297,805.15      | \$ 170,164.75  | 62    | \$ 169,552.83      | \$ 96,880.94   |
| 3     | \$ 295,985.29      | \$ 169,124.88  | 63    | \$ 167,050.03      | \$ 95,450.83   |
| 4     | \$ 294,155.75      | \$ 168,079.47  | 64    | \$ 164,533.90      | \$ 94,013.11   |
| 5     | \$ 292,316.46      | \$ 167,028.49  | 65    | \$ 162,004.37      | \$ 92,567.73   |
| 6     | \$ 290,467.37      | \$ 165,971.92  | 66    | \$ 159,461.38      | \$ 91,114.65   |
| 7     | \$ 288,608.44      | \$ 164,909.72  | 67    | \$ 156,904.84      | \$ 89,653.84   |
| 8     | \$ 286,739.61      | \$ 163,841.86  | 68    | \$ 154,334.69      | \$ 88,185.24   |
| 9     | \$ 284,860.83      | \$ 162,768.32  | 69    | \$ 151,750.85      | \$ 86,708.83   |
| 10    | \$ 282,972.04      | \$ 161,689.06  | 70    | \$ 149,153.25      | \$ 85,224.56   |
| 11    | \$ 281,073.20      | \$ 160,604.05  | 71    | \$ 146,541.82      | \$ 83,732.38   |
| 12    | \$ 279,164.24      | \$ 159,513.27  | 72    | \$ 143,916.49      | \$ 82,232.25   |
| 13    | \$ 277,245.12      | \$ 158,416.68  | 73    | \$ 141,277.17      | \$ 80,724.14   |
| 14    | \$ 275,315.78      | \$ 157,314.25  | 74    | \$ 138,623.80      | \$ 79,207.99   |
| 15    | \$ 273,376.17      | \$ 156,205.95  | 75    | \$ 135,956.31      | \$ 77,683.78   |
| 16    | \$ 271,426.23      | \$ 155,091.74  | 76    | \$ 133,274.60      | \$ 76,151.44   |
| 17    | \$ 269,465.90      | \$ 153,971.61  | 77    | \$ 130,578.62      | \$ 74,610.95   |
| 18    | \$ 267,495.14      | \$ 152,845.51  | 78    | \$ 127,868.28      | \$ 73,062.25   |
| 19    | \$ 265,513.88      | \$ 151,713.41  | 79    | \$ 125,143.51      | \$ 71,505.31   |
| 20    | \$ 263,522.07      | \$ 150,575.28  | 80    | \$ 122,404.23      | \$ 69,940.07   |
| 21    | \$ 261,519.66      | \$ 149,431.10  | 81    | \$ 119,650.36      | \$ 68,366.51   |
| 22    | \$ 259,506.58      | \$ 148,280.82  | 82    | \$ 116,881.83      | \$ 66,784.56   |
| 23    | \$ 257,482.78      | \$ 147,124.41  | 83    | \$ 114,098.56      | \$ 65,194.19   |
| 24    | \$ 255,448.21      | \$ 145,961.85  | 84    | \$ 111,300.46      | \$ 63,595.34   |
| 25    | \$ 253,402.80      | \$ 144,793.10  | 85    | \$ 108,487.47      | \$ 61,987.99   |
| 26    | \$ 251,346.50      | \$ 143,618.12  | 86    | \$ 105,659.49      | \$ 60,372.08   |
| 27    | \$ 249,279.25      | \$ 142,436.89  | 87    | \$ 102,816.46      | \$ 58,747.56   |
| 28    | \$ 247,200.99      | \$ 141,249.36  | 88    | \$ 99,958.29       | \$ 57,114.39   |
| 29    | \$ 245,111.67      | \$ 140,055.51  | 89    | \$ 97,084.89       | \$ 55,472.52   |
| 30    | \$ 243,011.21      | \$ 138,855.31  | 90    | \$ 94,196.20       | \$ 53,821.91   |
| 31    | \$ 240,899.58      | \$ 137,648.71  | 91    | \$ 91,292.13       | \$ 52,162.51   |
| 32    | \$ 238,776.70      | \$ 136,435.69  | 92    | \$ 88,372.59       | \$ 50,494.28   |
| 33    | \$ 236,642.52      | \$ 135,216.21  | 93    | \$ 85,437.50       | \$ 48,817.16   |
| 34    | \$ 234,496.97      | \$ 133,990.24  | 94    | \$ 82,486.79       | \$ 47,131.11   |
| 35    | \$ 232,339.99      | \$ 132,757.74  | 95    | \$ 79,520.36       | \$ 45,436.09   |
| 36    | \$ 230,171.53      | \$ 131,518.67  | 96    | \$ 76,538.13       | \$ 43,732.03   |
| 37    | \$ 227,991.53      | \$ 130,273.01  | 97    | \$ 73,540.03       | \$ 42,018.91   |
| 38    | \$ 225,799.91      | \$ 129,020.71  | 98    | \$ 70,525.96       | \$ 40,296.66   |
| 39    | \$ 223,596.63      | \$ 127,761.75  | 99    | \$ 67,495.84       | \$ 38,565.24   |
| 40    | \$ 221,381.61      | \$ 126,496.08  | 100   | \$ 64,449.59       | \$ 36,824.60   |
| 41    | \$ 219,154.80      | \$ 125,223.67  | 101   | \$ 61,387.11       | \$ 35,074.69   |
| 42    | \$ 216,916.13      | \$ 123,944.49  | 102   | \$ 58,308.33       | \$ 33,315.46   |
| 43    | \$ 214,665.53      | \$ 122,658.49  | 103   | \$ 55,213.15       | \$ 31,546.87   |
| 44    | \$ 212,402.96      | \$ 121,365.65  | 104   | \$ 52,101.49       | \$ 29,768.85   |
| 45    | \$ 210,128.33      | \$ 120,065.92  | 105   | \$ 48,973.26       | \$ 27,981.37   |
| 46    | \$ 207,841.60      | \$ 118,759.27  | 106   | \$ 45,828.37       | \$ 26,184.37   |



|    |               |               |     |              |              |
|----|---------------|---------------|-----|--------------|--------------|
| 47 | \$ 205,542.68 | \$ 117,445.67 | 107 | \$ 42,666.74 | \$ 24,377.80 |
| 48 | \$ 203,231.53 | \$ 116,125.06 | 108 | \$ 39,488.27 | \$ 22,561.62 |
| 49 | \$ 200,908.07 | \$ 114,797.43 | 109 | \$ 36,292.87 | \$ 20,735.76 |
| 50 | \$ 198,572.23 | \$ 113,462.73 | 110 | \$ 33,080.46 | \$ 18,900.17 |
| 51 | \$ 196,223.96 | \$ 112,120.91 | 111 | \$ 29,850.95 | \$ 17,054.82 |
| 52 | \$ 193,863.18 | \$ 110,771.96 | 112 | \$ 26,604.23 | \$ 15,199.64 |
| 53 | \$ 191,489.83 | \$ 109,415.82 | 113 | \$ 23,340.23 | \$ 13,334.57 |
| 54 | \$ 189,103.85 | \$ 108,052.46 | 114 | \$ 20,058.85 | \$ 11,459.58 |
| 55 | \$ 186,705.15 | \$ 106,681.84 | 115 | \$ 16,759.99 | \$ 9,574.60  |
| 56 | \$ 184,293.69 | \$ 105,303.92 | 116 | \$ 13,443.57 | \$ 7,679.59  |
| 57 | \$ 181,869.38 | \$ 103,918.66 | 117 | \$ 10,109.48 | \$ 5,774.48  |
| 58 | \$ 179,432.17 | \$ 102,526.03 | 118 | \$ 6,757.65  | \$ 3,859.23  |
| 59 | \$ 176,981.97 | \$ 101,125.98 | 119 | \$ 3,387.96  | \$ 1,933.78  |
|    |               |               | 120 | \$ 0.33      | \$ (1.92)    |